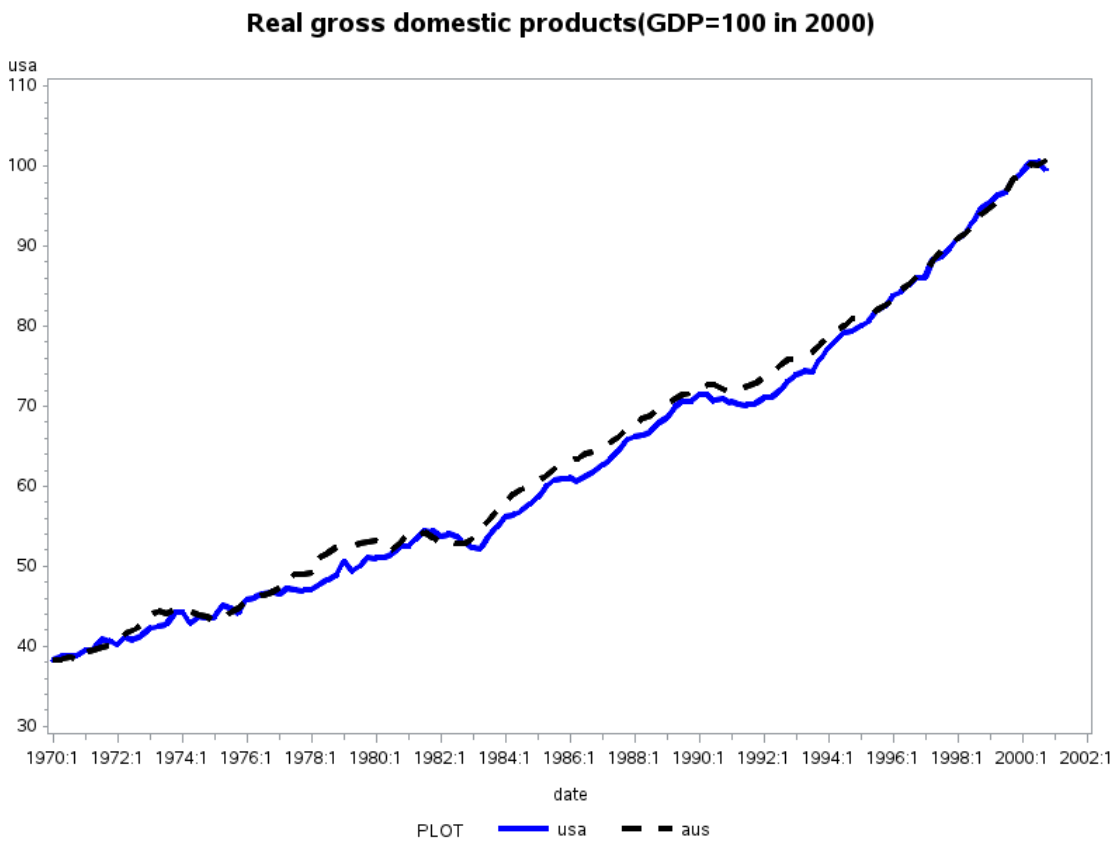
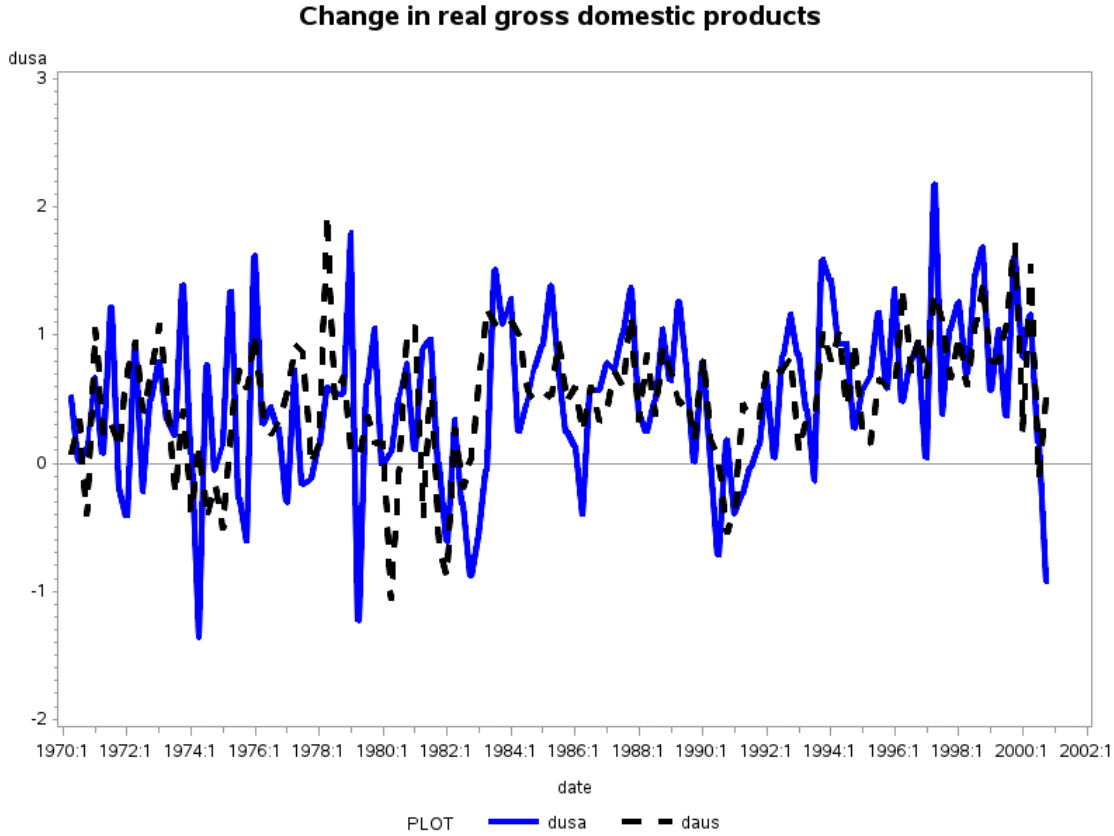
first 5 observations for GDP data

Obs	aus	usa	dusa	daus	date	year	qtr
1	38.3011	38.2355	.	.	1970:1	1970	1
2	38.3734	38.7551	0.5196	0.0723	1970:2	1970	2
3	38.7137	38.7706	0.0155	0.3403	1970:3	1970	3
4	38.2991	38.8948	0.1242	-0.4146	1970:4	1970	4
5	39.3615	39.5621	0.6673	1.0624	1971:1	1971	1





ADF test for real GDP: U.S.

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	time	DLAG_V	AR_V1	AR_V2	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.40749	0.49322	-1	0.009390041	-0.009772	0.055498	0.008058	121	-0.54091	2	1	0.98033
2	COV	0 Converged	AR_V	Intercept	0.40749	0.35039	.	0.004747157	-0.010468	0.010072	0.012368	121	-0.54091	2	1	0.98033
3	COV	0 Converged	AR_V	time	0.40749	0.00475	.	0.000075087	-0.000153	0.000114	0.000145	121	-0.54091	2	1	0.98033
4	COV	0 Converged	AR_V	DLAG_V	0.40749	-0.01047	.	-.000152977	0.000326	-0.000355	-0.000425	121	-0.54091	2	1	0.98033
5	COV	0 Converged	AR_V	AR_V1	0.40749	0.01007	.	0.000113520	-0.000355	0.009563	0.000317	121	-0.54091	2	1	0.98033
6	COV	0 Converged	AR_V	AR_V2	0.40749	0.01237	.	0.000144678	-0.000425	0.000317	0.009807	121	-0.54091	2	1	0.98033

ADF test for real GDP: Australia

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	time	DLAG_V	AR_V1	AR_V2	AR_V3	AR_V4	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.22477	0.66162	-1	0.010499	-0.016241	0.23183	0.19166	-.000460692	0.012585	119	-0.90827	2	1	0.95110
2	COV	0 Converged	AR_V	Intercept	0.22477	0.34331	.	0.004733	-0.010334	0.00632	0.00927	0.009073116	0.014246	119	-0.90827	2	1	0.95110
3	COV	0 Converged	AR_V	time	0.22477	0.00473	.	0.000072	-0.000149	0.00009	0.00012	0.000124313	0.000191	119	-0.90827	2	1	0.95110
4	COV	0 Converged	AR_V	DLAG_V	0.22477	-0.01033	.	-0.000149	0.000320	-0.00024	-0.00031	-.000307234	-0.000472	119	-0.90827	2	1	0.95110
5	COV	0 Converged	AR_V	AR_V1	0.22477	0.00632	.	0.000088	-0.000239	0.00877	-0.00139	-.001435638	0.000579	119	-0.90827	2	1	0.95110
6	COV	0 Converged	AR_V	AR_V2	0.22477	0.00927	.	0.000120	-0.000310	-0.00139	0.00937	-.001258384	-0.001022	119	-0.90827	2	1	0.95110
7	COV	0 Converged	AR_V	AR_V3	0.22477	0.00907	.	0.000124	-0.000307	-0.00144	-0.00126	0.009807813	-0.001701	119	-0.90827	2	1	0.95110
8	COV	0 Converged	AR_V	AR_V4	0.22477	0.01425	.	0.000191	-0.000472	0.00058	-0.00102	-.001701495	0.009819	119	-0.90827	2	1	0.95110

ADF test for differenced real GDP: U.S.

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.42514	0.40502	-1	-0.80796	-0.072489	121	-6.51002	1	1	.000084295
2	COV	0 Converged	AR_V	Intercept	0.42514	0.00756	.	-0.00790	0.003968	121	-6.51002	1	1	.000084295
3	COV	0 Converged	AR_V	DLAG_V	0.42514	-0.00790	.	0.01540	-0.007712	121	-6.51002	1	1	.000084295
4	COV	0 Converged	AR_V	AR_V1	0.42514	0.00397	.	-0.00771	0.008744	121	-6.51002	1	1	.000084295

ADF test for difference real GDP: Australia

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	AR_V2	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.23261	0.28175	-1	-0.53607	-0.23463	-0.013422	120	-4.62348	1	1	.000274905
2	COV	0 Converged	AR_V	Intercept	0.23261	0.00545	.	-0.00687	0.00457	0.002264	120	-4.62348	1	1	.000274905
3	COV	0 Converged	AR_V	DLAG_V	0.23261	-0.00687	.	0.01344	-0.00897	-0.004584	120	-4.62348	1	1	.000274905
4	COV	0 Converged	AR_V	AR_V1	0.23261	0.00457	.	-0.00897	0.01293	0.006602	120	-4.62348	1	1	.000274905
5	COV	0 Converged	AR_V	AR_V2	0.23261	0.00226	.	-0.00458	0.00660	0.008958	120	-4.62348	1	1	.000274905

regress Australian GDP on U.S. GDP

The AUTOREG Procedure

Dependent Variable	aus

regress Australian GDP on U.S. GDP

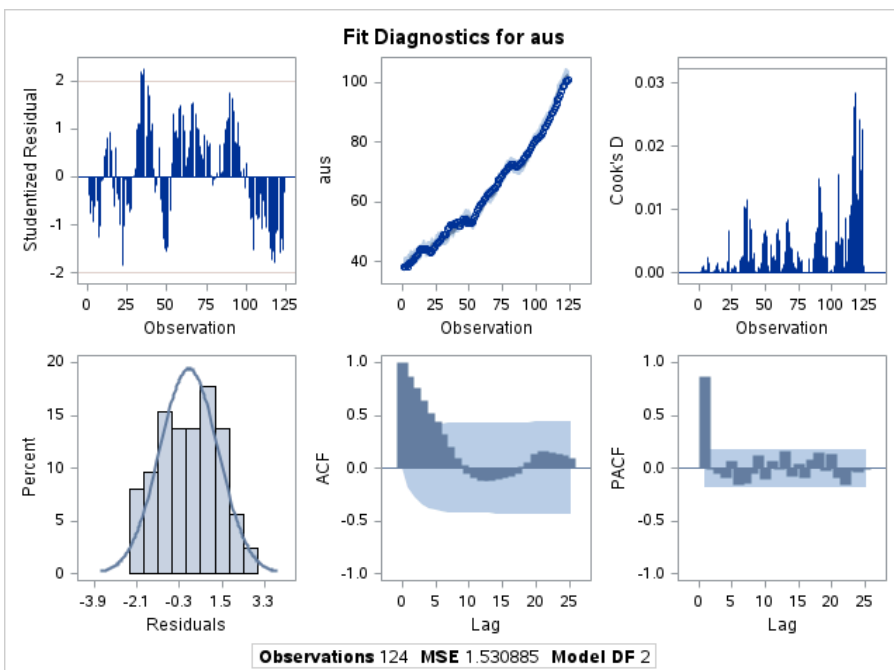
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	188.298895	DFE	123
MSE	1.53089	Root MSE	1.23729
SBC	408.517914	AIC	405.697632
MAE	1.04749852	AICC	405.730419
MAPE	1.73567485	HQC	406.843297
Durbin-Watson	0.2553	Total R-Square	0.9997
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
usa	1	1.0145	0.001706	594.79	<.0001

regress Australian GDP on U.S. GDP

The AUTOREG Procedure



test for cointegration

The AUTOREG Procedure

Dependent Variable	dehat

test for cointegration

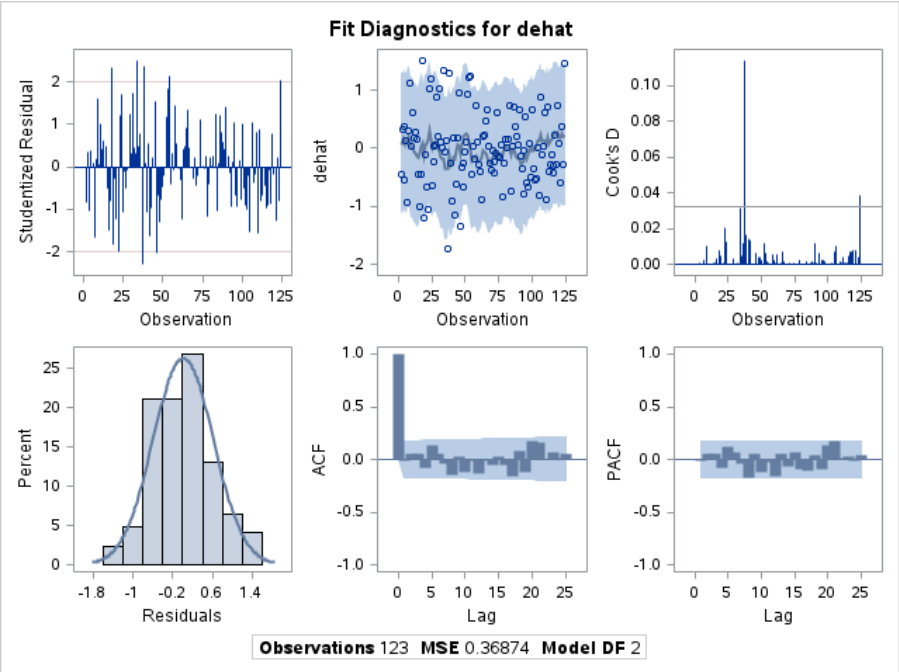
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	44.9863347	DFE	122
MSE	0.36874	Root MSE	0.60724
SBC	230.154517	AIC	227.342332
MAE	0.47976677	AICC	227.37539
MAPE	159.604839	HQC	228.484635
Durbin-Watson	1.9782	Total R-Square	0.0641
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
ehat1	1	-0.1279	0.0443	-2.89	0.0046

test for cointegration

The AUTOREG Procedure



estimate response in Australia to a cointegrating error

The AUTOREG Procedure

Dependent Variable	daus

estimate response in Australia to a cointegrating error

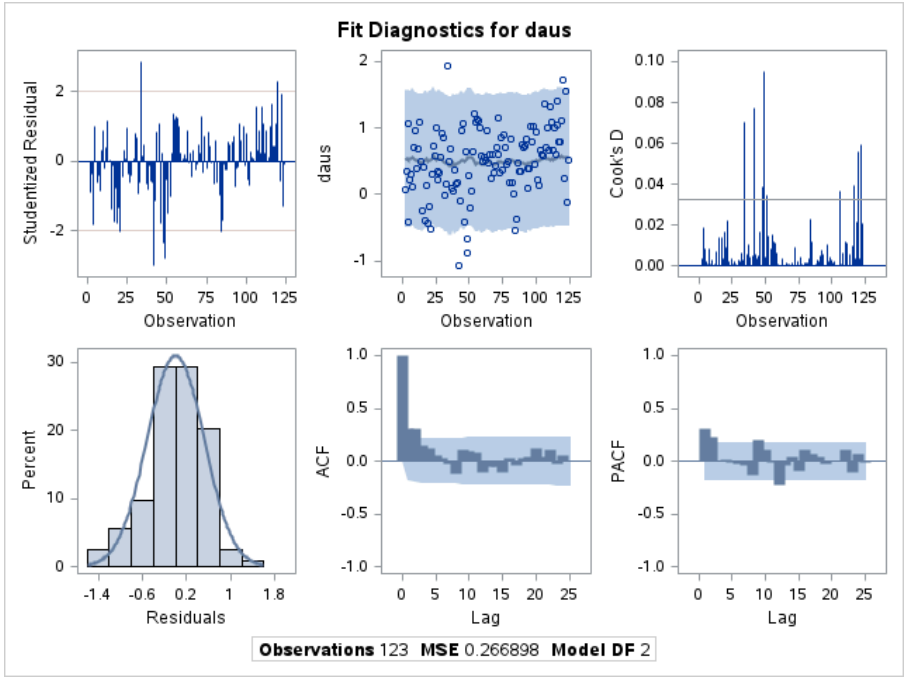
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	32.2947105	DFE	121
MSE	0.26690	Root MSE	0.51662
SBC	194.197697	AIC	188.573328
MAE	0.39378107	AICC	188.673328
MAPE	264.849196	HQC	190.857933
Durbin-Watson	1.3677	Total R-Square	0.0049

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.5105	0.0467	10.92	<.0001
ehat1	1	-0.0292	0.0378	-0.77	0.4407

estimate response in Australia to a cointegrating error

The AUTOREG Procedure



estimated response in U.S. to a cointegrating error

The AUTOREG Procedure

Dependent Variable	dusa

estimated response in U.S. to a cointegrating error

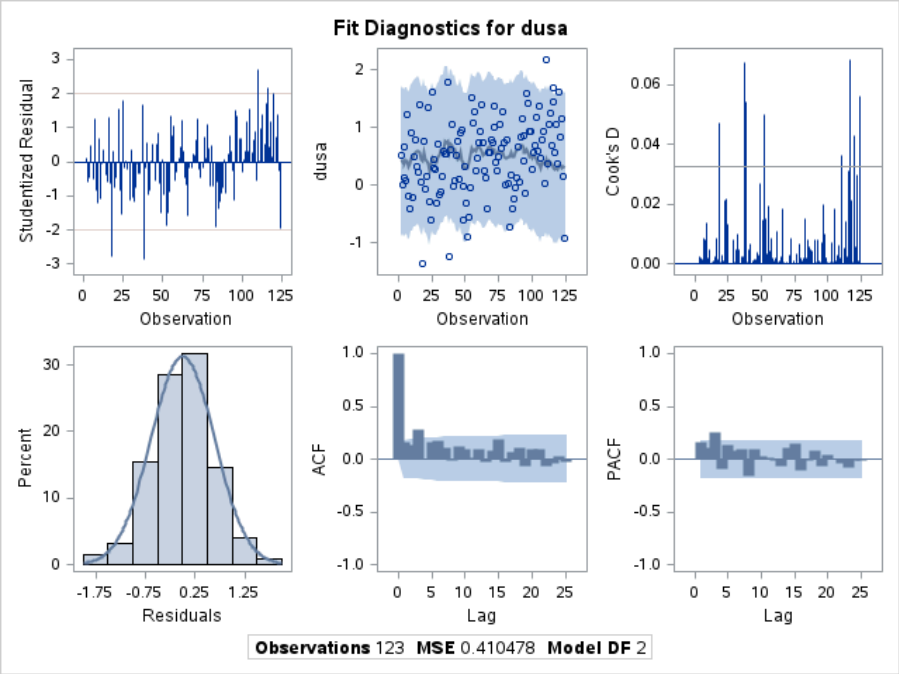
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	49.6678896	**DFE**	121
MSE	0.41048	**Root MSE**	0.64069
SBC	247.143685	**AIC**	241.519316
MAE	0.50717838	**AICC**	241.619316
MAPE	300.022884	**HQC**	243.803921
Durbin-Watson	1.6399	**Total R-Square**	0.0350

Parameter Estimates					
Variable	**DF**	**Estimate**	**Standard Error**	**t Value**	**Approx Pr > |t|**
Intercept	1	0.4895	0.0580	8.44	<.0001
ehat1	1	0.0982	0.0469	2.10	0.0382

estimated response in U.S. to a cointegrating error

The AUTOREG Procedure



estimated VEC model

The VARMAX Procedure

Number of Observations	124
Number of Pairwise Missing	0

Simple Summary Statistics

Variable	Type	N	Mean	Standard Deviation	Min	Max
aus	Dependent	124	63.73438	17.59423	38.29910	100.72100
usa	Dependent	124	62.72528	17.65155	38.23550	100.60400

Dickey-Fuller Unit Root Tests

Variable	Type	Rho	Pr < Rho	Tau	Pr < Tau
aus	Zero Mean	0.99	0.9087	6.37	0.9999
	Single Mean	1.10	0.9898	2.57	0.9999
	Trend	-0.68	0.9911	-0.27	0.9908
usa	Zero Mean	0.99	0.9084	6.47	0.9999
	Single Mean	1.20	0.9912	2.66	0.9999
	Trend	-1.29	0.9836	-0.58	0.9784

estimated VEC model

The VARMAX Procedure

Type of Model	VECM(2)
Estimation Method	Maximum Likelihood Estimation
Cointegrated Rank	1

Model Parameter Estimates

Equation	Parameter	Estimate	Standard Error	t Value	Pr > |t|	Variable
D_aus	CONST1	0.18021	0.14959	1.20	0.2307	1
	AR1_1_1	0.03726	0.03123	1.19	0.2353	aus(t-1)
	AR1_1_2	-0.03495	0.02930	-1.19	0.2353	usa(t-1)
	AR2_1_1	0.22812	0.10228	2.23	0.0276	D_aus(t-1)
	AR2_1_2	0.06449	0.07775	0.83	0.4085	D_usa(t-1)
D_usa	CONST2	-0.27129	0.18264	-1.49	0.1401	1
	AR1_2_1	0.13646	0.03813	3.58	0.0005	aus(t-1)
	AR1_2_2	-0.12802	0.03577	-3.58	0.0005	usa(t-1)
	AR2_2_1	0.20644	0.12488	1.65	0.1010	D_aus(t-1)
	AR2_2_2	-0.00219	0.09493	-0.02	0.9816	D_usa(t-1)

Alpha and Beta Parameter Estimates

Equation	Parameter	Estimate	Standard Error	t Value	Pr > |t|	Variable

Alpha and Beta Parameter Estimates						
Equation	Parameter	Estimate	Standard Error	t Value	Pr > |t|	Variable
D_aus	ALPHA1_1	0.03726	0.03123	1.19	0.2353	Beta[,1]*_DEP_(t-1)
	BETA1_1	1.00000				aus(t-1)
D_usa	ALPHA2_1	0.13646	0.03813	3.58	0.0005	Beta[,1]*_DEP_(t-1)
	BETA2_1	-0.93817				usa(t-1)

Covariance Parameter Estimates				
Parameter	Estimate	Standard Error	t Value	Pr > |t|
COV1_1	0.23547	0.03015	7.81	<.0001
COV1_2	0.11195	0.02793	4.01	0.0001
COV2_2	0.35101	0.04494	7.81	<.0001

Covariances of Innovations		
Variable	aus	usa
aus	0.23547	0.11195
usa	0.11195	0.35101

Log-likelihood	40.11162

Information Criteria	
AICC	-58.2413
HQC	-48.8342
AIC	-60.2232
SBC	-32.183
FPEC	0.079948